

Good morning.

U O X O G H U L B S O L I F B B W F L R V Q Q P R N G K S S O
T W T Q S J Q S S E K Z Z W A T J K L U D I A W I N F B N Y P
V T T M Z F P K W G D K Z X T J C D I G K U H U A U E K C A R

I think the algorithm used to create K4 combined the algorithms used in K1 through K3. I believe the message was first encrypted with a substitution cipher. That encryption was then transposed using a algorithm possibly inspired by the Mengenlehreuhr clock. The text as it currently stands looks nothing like a simple aristocrat and yet that is kind of what it is. For me the first indicator is the clustering of letters. All of the R's are on the right end. All of the O's are in the top line. There are six sets of double letters. Half of the U's are within one letter of each other in the bottom line. These are statistically important letters. The O's are 5% of the text, the U's are 6%.

Berlin clock maps to the bottom left corner. North East begins just below and to the left of OBKR. These two clues are integral to the transposition. Clock points to how the Berlin Clock works. Northeast is Southwest of OBKR.

Starting at the O of OBKR and using the "second" sequence of the clock count 5 letters and shift the 5th letter to the upper position. Here's a thousand words ...

Red pass #1, Green pass #2, Blue pass #3, Violet pass #4, Pass #5, (20 characters)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
1 O B K R
1 2 3 4
2 U O X O G H U L B S O L I F B B W F L R V Q Q P R N G K S S O
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
E A S T N O R T H E A S T
3 T W T Q S J Q S S E K Z Z W A T J K L U D I A W I N F B N Y P
36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66
B E R
4 V T T M Z F P K W G D K Z X T J C D I G K U H U A U E K C A R
67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97
L I N C L O C K

U H O B
1 2 3 4
V N O S E A U I Y M W X I U C K O B F L P S T S Z K A B T P K
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
C U E O O U L W Q G T J K T D N P Z G T G A A R G S B R R S Q
36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66
S W L W N T K Z D H K B X L I F Q K W Q Z J I F V F D J K U R
67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97

How does the reversal work, from the bottom or the top?